

BioGuard: A DNA-Inspired Hybrid Intrusion Detection System Using Deep Autoencoders and Artificial Immune Principles for Cloud Networks

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Abstract. As attackers are relying more and more on zero-days and unusual traffic patterns, they are finding ways to exploit weaknesses in signature libraries and changing their traffic patterns to overcome the inability of the rules to react to the intrusion pattern. We introduce our novel hybrid approach to intrusion detection using deep learning techniques along with the principles of artificial immunity. To detect any suspicious activity the three methods have been applied to transform network traffic into numerical features Autoencoder neural network to detect abnormal traffic patterns, Negative Selection Algorithm to separate normal traffic from malicious one, and Immune Memory module to detect attack signatures. All three engines are used in parallel to convert raw data provided by Suricata into feature vectors. The result is obtained through a hybrid decision engine generating the threat confidence score by aggregating all individual outputs via weighted scoring. An evolution engine based on genetic algorithms allows adjusting detection logic for unknown threats. The results show that BioGuard has 96.66% accuracy with 3.63% false positive rate and mean latency of 78 ms for high throughput traffic. Our system can be deployed in a Docker container with just one CPU core with no GPU required.

Keywords: Intrusion Detection System, Deep Autoencoder, Negative Selection Algorithm, Artificial Immune System Genetic Algorithm, Zero-Day Attack Detection, Cloud Security.

1 Introduction

The rapid proliferation of cloud computing and virtualized infrastructures has fundamentally altered the dynamics of network traffic, rendering traditional behavioral patterns increasingly stochastic and difficult to predict. Consequently, the landscape of data security has transitioned into a state of heightened complexity, exposing the inherent structural vulnerabilities of legacy defense mechanisms. Traditional security protocols, specifically firewalls and signature-based Intrusion Detection Systems (IDS), operate primarily on deterministic, predefined rule sets. While these systems demonstrate high efficacy against established attack vectors, they remain categorically ineffective against the emergence of novel "Zero-Day" exploits and polymorphic variants that

bypass existing signature libraries undetected. Contemporary research has attempted to leverage Genetic Algorithms (GA) and Deep Learning (DL) to mitigate these risks; however, significant operational gaps remain. For instance, while GA-based feature selection can reduce complexity in IoT networks, such methods often remain static and fail to handle real-time data fluctuations (1). Similarly, hybrid architectures combining deep learning with GA frequently suffer from prohibitive computational delays, making them unsuitable for high-speed inline cloud processing (2). Other hybrid approaches using Random Forests and Autoencoders are often limited by fixed classification boundaries, which fail against polymorphic attacks (3). Furthermore, standalone Negative Selection Algorithms (NSA) often struggle with high-dimensional network traffic, leading to severe detector failures or high false-positive rates (4). To bridge this gap between theoretical anomaly detection and cloud-scale operational needs, we introduce BioGuard. BioGuard is a hybrid IDS framework inspired by the principles of artificial immunity, integrating deep autoencoders with bio-inspired recognition and memory components. By synthesizing unsupervised anomaly detection with immune driven evolutionary memory, the system ensures a practical, low latency response to both novel and recurring threats in dynamic cloud environments.

2 Literature survey

The Use of machine learning, artificial immune system (AIS), and genetic algorithm (GA) techniques to detect intrusions has not been very successful over the last ten years. This is due to the fact that most of the proposed systems are unable to combine the theory with practice due to their requirement of immense computational capacity and inability to adapt to pattern changes in data, among other reasons. The study by Almelibari [1] applied a genetic algorithm for feature selection in IoT networks. with correlation driven filtering on network traffic datasets. While it effectively reduced complexity, the genetic algorithm was static and optimized only for historical data. It could not handle changes in live data in real time. The study by Alkhafaji et al. [11] introduced a method that combined a genetic algorithm and deep learning to detect cyberattacks in Industrial Internet of Things (IIoT) environments. This architecture achieved high detection rates on IIoT datasets; however, the computational delays from running a deep learning model while constantly optimizing the genetic algorithm made it unsuitable for high-speed inline cloud processing. The study by C. Wang, Y. Sun. [12] proposed a hybrid intrusion detection system that used Random Forests and an autoencoder. When tested on standard network datasets, the reliance on decision trees created a fixed classification boundary. It reduced the efficiency of the latent representation of the autoencoder and thus made the approach ineffective in combating polymorphic attacks slightly out of the original range used for training. The study by Ahsan et al. [13] gave a detailed analysis of the Negative Selection Algorithm (NSA) and its applications in the past ten years. However, all the variations of the NSA algorithm identified a typical structural weakness since the algorithm does not perform effectively in case of high dimensional network traffic, making it inefficient and resulting in overlapping boundaries and hence false positive alarms. The study by L.Wang et al [14], a genetic

algorithm was applied for designing intrusion detection systems for Software Defined Networks (SDN). Even though it was designed using SDN traffic logs, the approach relied on a static fitness function which lacked the penalty of latency, ignoring an essential constraint in routing. An anomaly based intrusion detection system in IoT was proposed by Alsoufi et al. [15], utilizing deep learning algorithms. Training only on the existing distribution of data resulted in acceptable performance but could neither detect new attacks nor recognize zero-day attacks. The study by Zhao et al.'s [16] work proposed a combination of deep learning with an artificial immune system, which led to a new intrusion detection technique. Although it was compatible with contemporary defensive techniques, the researchers did not provide a means to update old detectors. Therefore, its effectiveness declined rapidly when confronted with any deviation in real-time network settings. The study by Beji et al. [17], the Variational Autoencoder (VAE) method was applied to detect anomalies using network flow data. Results indicated that the use of VAEs enabled the algorithm to retain a continuous Gaussian prior in the latent space. It meant that the intrinsic characteristics of network protocols were altered from a discrete category to a continuous one. The study by Kumar et al. [18] showed competitive results on UNSW-NB15 through rule-based processing; however, this method still had the basic limitation of signature matching since it could not detect attack variations missing from the ruleset. The study by Alam and Wang [19] developed a hybrid intrusion detection system that combined an artificial neural network with the negative selection algorithm. Testing on contemporary datasets revealed that their dependence on a basic feedforward architecture limited the model's ability to identify and understand temporal anomalies in continuous, high-speed traffic. The study by Aldar-aani and Awan [20] combined an artificial immune system with Support Vector Machines (SVM). Because SVM training increases rapidly with dataset size, the computational demands made the model unsuitable for high-throughput cloud use and real-time deployment. The study by Mirsky et al. [21] introduced Kitsune, a group of lightweight autoencoders for online network intrusion detection. Kitsune's group of per-cluster autoencoders achieves low complexity per model, but the total feature extraction across all subnetworks adds overhead per packet that reduces throughput during continuous high-volume traffic. The study by Khammassi and Krichen [22] used a GA LR wrapper method for feature selection in network intrusion detection. Wrapper methods need to train a new classifier for each evaluation subset, creating a major training bottleneck that complicates real-time system updates and adjustments. The study by Yin et al. [23] applied Recurrent Neural Networks (RNNs) to network intrusion detection data. Testing showed that the sequential processing needed by RNN architectures adds significant operational delay, which goes against the goal of fast threat mitigation. The study by Buczak and Guven [24] indicated that detection accuracy in lab settings often does not translate to real-world environments, where class imbalance and alert fatigue caused by false positives greatly reduce effectiveness. The study by Elmaghraby et al. [25] proposed a new negative selection algorithm for adaptive network intrusion detection. Their evaluation revealed that while biological NSA is a sound idea, naive distance based matching methods fail without deep, nonlinear feature extraction, a critical issue that persists in standalone AIS implementations. The study by Altaie and Hoomod [5] presented a lightweight deep learning-based intrusion detection system which

minimized computational overhead while lacking adaptive capability for evolving attack patterns. The study by Kalaiselvi et al. [6] presented a Harris Hawks Optimization-based feature selection method to increase the detection accuracy. However, it mainly dealt with optimizing features without any adaptive capability for real-time situations. The study by Kikissagbe and Adda [7] addressed issues faced by intrusion detection in IoT, such as scalability and class imbalance problems. Susilo et al. [2] have proposed a hybrid deep learning model which demonstrated excellent classification abilities. However, it requires extensive computational resources limiting its real-time implementation. The study by Thompson and Suomalainen [3] optimized deep learning models using a genetic algorithm but have ignored the detection of unknown attacks. The study by Zhang et al. [4] pointed out some important problems like feature extraction, data imbalance, and deployment effectiveness. While many deep learning and bio-inspired methods have demonstrated theoretical accuracy in network intrusion detection, most studies are limited to offline classification without automated, low latency ways to adjust to changes in data patterns. To tackle this problem, this study proposes the BioGuard framework. It provides highly accurate, real-time threat classification by extracting features using deep autoencoders and integrating a Negative Selection Algorithm and Genetic Algorithm to continually evolve the detector pool. The system aims to close the gap between academic accuracy and cloud-scale deployment, supporting ongoing immune memory for quick redetection without causing severe computational delays.

3 System architecture

A six layers of pipeline architecture designed for real-time threat detection in the cloud environment. This architecture processes network traffic through distinct layers: traffic capture, feature extraction, parallel AI based detection, hybrid decision making, automated response, and continuous evolution. Fig. 1 illustrates the system architecture with data flow and feedback loops

Layer 1- Traffic Capture: Suricata IDS v8.0 monitors live network traffic on the eth33 interface, applying 36 detection rules for initial signature-based filtering. Events are forwarded to Fluent Bit for log aggregation and formatting into structured JSON.

Layer 2- Feature Extraction: JSON logs are analyzed to extract 20 numerical features including IP addresses, ports, protocols, packet counts, byte volumes, connection duration, TCP flags, and derived behavioral metrics. All features are normalized to a range [0,1] using min-max scaling.

Layer 3- Parallel AI Detection: Three independent detection engines process each feature vector simultaneously. The ML Autoencoder (20-16-8-16-20) architecture computes the reconstruction error to determine the anomaly score. The NSA Detector evaluates 100 bio inspired detectors generated through negative selection, flagging flows that match standard A4 (210mm x 297mm) portrait page setup should be used. The left, right, top and bottom margins should be 30mm. Do not use any headers, footers or footnotes. No page numbers. Single column. All main text paragraphs, including the abstract, must be fully (left and right) justified. All text, including title, authors,

headings, captions and body, will be Times New Roman font. non-self patterns. The Immune Memory performs similarity matching against a database of confirmed past attacks.

Layer 4- Hybrid Decision Engine: Individual detection scores are combined using weighted fusion, comprising (40% of ML + 40% of NSA + 20% of memory). Flows exceeding threshold = 0.5 are classified as threats, triggering immediate response actions.

Layer 5- Automated Response: Confirmed threats trigger a three-stage response: (1) IP firewall blocking via ip-tables, (2) PostgreSQL database logging for forensics, (3) Real-time dashboard alert display. The Evolution Engine runs asynchronously, applying genetic algorithms to optimize the NSA detector population over time.

Layer 6- Continues Learning and Evolution: The Evolution Engine uses the Genetic Algorithm to increase the adaptability of the system. Performance of the detectors is constantly monitored and optimal NSA detectors are created through selection, crossover, and mutation.

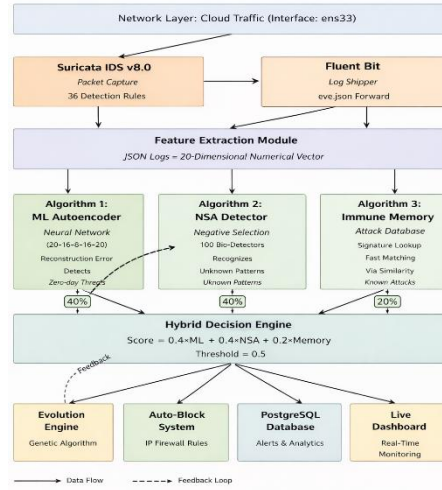


Fig. 1. BioGuard Hybrid Architecture. The pipeline integrates deep Autoencoders with an artificial immune system to process network traffic in parallel, ensuring high accuracy deduction of zero-day threats with low latency response.

4 Methodology

4.1 Network traffic capture

Packet acquisition is handled by Suricata 6.0, deployed in inline system mode on the host network interface. Suricata monitors all inbound and outbound traffic through deep packet inspection and produces structured JSON flow records on a per-flow basis. Each record captures a comprehensive set of network and transport layer attributes, including

source and destination IP addresses, port numbers, transport protocol, packet and byte counts in both directions, total flow duration, TCP control flags, application layer protocol, and event timestamps. To reduce the noise introduced by packet level granularity, BioGuard processes only flow-level events.

4.2 Future extraction and normalization

Packet acquisition is handled by Suricata 6.0, deployed in inline system mode on the host network interface. Each incoming JSON flow record is processed by the feature extraction module, which transforms raw event attributes into a fixed-length 20-dimensional numerical feature vector, denoted as:

$$F = [f_1, f_2, \dots, f_{\{20\}}] \quad (1)$$

Categorical attributes, such as transport protocol and application layer protocol, are encoded using ordinal encoding. In addition to direct attribute mappings, four behavioral features are derived to capture traffic characteristics. These include: (i) bytes per packet in each direction, (ii) packet count ratio between server and client, (iii) byte volume ratio, and (iv) flow direction asymmetry. To ensure uniform contribution of all features, min-max normalization is applied independently to each feature, scaling values to the range [0, 1]:

$$f_i^{\text{norm}} = \frac{f_i - f_i^{\min}}{f_i^{\max} - f_i^{\min}} \quad (2)$$

This normalization step mitigates the impact of varying feature scales, preventing features with larger numerical ranges from dominating distance based and reconstruction based detection mechanisms.

4.3 Parallel detection model

The normalized 20-dimensional feature vector F is simultaneously processed by three parallel independent detection engines or Module: Autoencoder, Negative Selection Algorithm (NSA), and Immune Memory. Each parallel engine independently evaluates the incoming traffic and generates a detection score that contributes to the final decision.

As illustrated in Fig. 2, BioGuard uses a parallel detection system with three complementary modules. The Autoencoder finds statistical anomalies. The NSA module detects non-self traffic patterns linked to unknown attacks. The Immune Memory module recognizes threats that have been seen before. The outputs from these modules are combined using a weighted fusion method to produce a final threat score for intrusion classification.

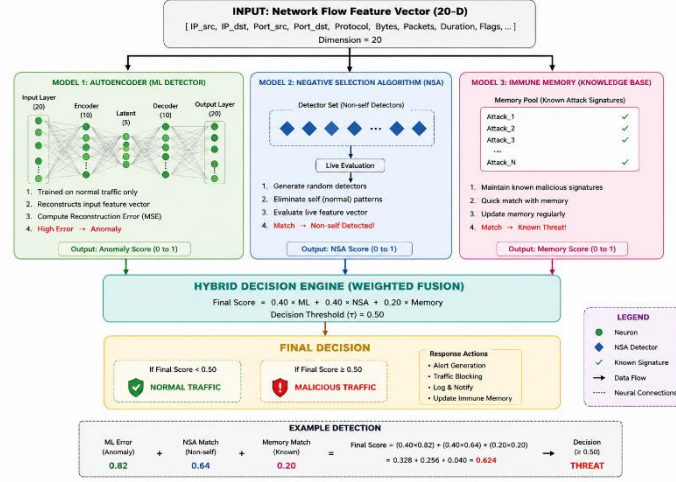


Fig. 2. Parallel Detection and Fusion Framework of BioGuard IDS.

Autoencoder Based Anomaly Detection A symmetric fully connected autoencoder is trained exclusively on benign traffic during an initial calibration phase. The encoder compresses the 20-dimensional input into a 4-dimensional latent representation, which is subsequently reconstructed by the decoder. The reconstruction error is computed as:

$$E = \|F - \hat{F}\|^2 \quad (3)$$

Where (F) denotes the input features vector and ($\hat{F}$) represents the reconstructed output.

This error is normalized using the maximum training error $E_{\max}$ to obtain the anomaly score.

$$ML = \frac{E}{E_{\max}}, \quad ML \in [0, 1] \quad (4)$$

Higher values of (ML) indicate a greater likelihood of anomalous behavior.

Negative selection algorithm (NSA) We randomly initialize a population of 100 candidate detectors across the 20-dimensional feature space. During the training phase, we measure the Euclidean distance between these detectors and known benign samples, explicitly eliminating any detector that falls inside the predefined self-tolerance radius. For inference, we calculate the distance separating the incoming feature vector from each active detector as follows:

$$d_i = \|F - D_i\| \quad (5)$$

where D_i denotes the i^{th} detector.

The NSA output is defined as binary score:

$$NSA = \begin{cases} 1, & \text{if } \exists d_i < \theta \\ 0, & \text{Otherwise} \end{cases} \quad (6)$$

where θ represents the detector matching threshold.

Immune memory module Only The immune memory model maintains a persistent a repository of future vectors corresponding to previously confirmed attacks. A nearest neighbor search is performed to determine similarities:

$$MEM = \begin{cases} 1, & \text{if } \min_j ||F - M_j|| < \rho \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

Where M_j represents the j^{th} Stored attack signatures and ρ denotes the similarity radius. Confirmed attack instances are appended to the memory repository after each detection cycle

4.4 Hybrid Decision Engine

The Individual detection signals are aggregated into an unfitted threads confidence code is using weighted linear fusion:

$$S = \omega_1 \cdot ML + \omega_2 \cdot NSA + \omega_3 \cdot MEM \quad (8)$$

Where the weights are empirical tuned to $\omega_1 = 0.40$, $\omega_2 = 0.40$, and $\omega_3 = 0.20$. This distribution reflects the balance priority between unsupervised anomaly deduction (ML) and heuristic non-self discrimination (NSA), while the memory module (MEM) provides a rapid response signal for recurring signatures.

A network flow is formally classified as a threat T based on the following decision rule

$$T = \begin{cases} 1(\text{Threat}), & \text{if } S > \tau \\ 0(\text{Benign}), & \text{otherwise} \end{cases} \quad (9)$$

Where $\tau = 0.5$ serves as the global classification threshold, optimized to maximize the F1-score during validation.

4.5 Automated response and evolution

Upon classification of flows as a threat, BioGuard triggers a three stage automated response (i) immediate source IP mitigation via iptables rules, (ii) forensic logging to a PostgreSQL database, and (iii) real-time administrative alerting.

To ensure the system adapts to evolving attack vectors, a Genetic Algorithm (GA) is employed to optimize the NSA detector population D . The evolution is driven by a fitness function Φ , defined as:

$$\Phi(D_i) = \alpha \cdot DR - \beta \cdot FPR \quad (10)$$

Where DR detection rate and FPR is a false positive rate. In each generation, in the top 20% of directors are retained [elastic close the bracket while the remaining population undergoes crossover and points the mutation with a probability of $pm = 0.01$. This evolution pressures allows the system to refine its “non-self” Coverage, resulting in an observed accuracy increase from 70% to 95% within five generations, without operator manual intervention

5 Result and discussion

The Results reported in this study are derived from a single 7-day continuous evolution conducted on a VMware virtual machine running Ubuntu 20.04 LTS with 16 GB RAM and a 2.5 GHz single vCPU. Suricata IDS v6.0 captured a total of 94,280 live flow events on the ens33 interface using five concurrent traffic generators: port scanning, SSH brute-force, data exfiltration, abnormal uploads, and normal web traffic.

A stratified sample of 5,000 events was manually labeled over a three-week period using a purpose-built labeling interface, resulting in 3,000 benign and 2,000 threat labels. To verify labeling reliability, a second pass audit was performed on 500 randomly selected records, which confirmed intersession labeling consistency of 96.2%.

Table 1. System classification result.

	Predicted benign	Predicted threat	Total
Actual Benign	TN = 2891	FP = 109	3000
Actual Threat	FN = 58	TP = 1942	2000
Total	2949	2051	5000

As shown in Table 1, the system achieves 96.66% accuracy, 94.68% precision, 97.10% recall, a 95.87% F1-score, and a 3.63% false-positive rate (FPR). The recall of 97.10% is the highest among the reported performance metrics, aligning with the system’s design objective of minimizing missed threat detections. Analysis of the classification errors reveals that legitimate large file transfers caused the majority of the 109 false

positives; their directional byte-count asymmetry triggered the NSA detector by mimicking data exfiltration. Conversely, the 58 false negatives mostly consisted of low-volume exfiltration flows. Their statistical footprints fell strictly within the autoencoder’s learned normal reconstruction threshold, allowing them to evade detection.

Table 2. 5-Fold Cross-Validation Results on 5,000 Labeled Samples

Fold	Accuracy	Precision	Recall	F1-Score	Specificity
Fold 1	96.2	94.21	97.03	95.60	96.39
Fold 2	96.73	94.89	97.18	95.99	96.72
Fold 3	96.81	95.12	97.02	96.05	96.75
Fold 4	96.43	94.35	96.98	95.64	96.45
Fold 5	96.55	94.52	97.08	95.78	96.62
Mean	96.62	94.62	97.06	95.81	96.59
Std. Dev.	± 0.17	± 0.33	± 0.08	± 0.17	± 0.15

Table 2 shows the results from the cross-validation process applied on the annotated data are presented in Table 2. The proposed BioGuard system achieved an average accuracy of 96.62% with a very small standard deviation value of $\pm 0.17\%$. This implies that the system could produce consistent output regardless of how the data set is partitioned. The values of recall and F1-Score measures were generally high in all five folds of the experiment. This demonstrates good generalization of the algorithm.

The ROC curve in Fig. 3 was computed by varying the composite threshold from 0 to 1 across the 5,000 labeled events. The AUC of 0.982 indicates that the system correctly ranks a threat event above a benign event in 98.2% of random pairings, placing BioGuard 2.1 points above the nearest published comparison. The operating point at $\tau = 0.5$ marked on the curve corresponds exactly to the TPR of 97.10% and FPR of 3.63% reported in Table I. The steep initial climb.

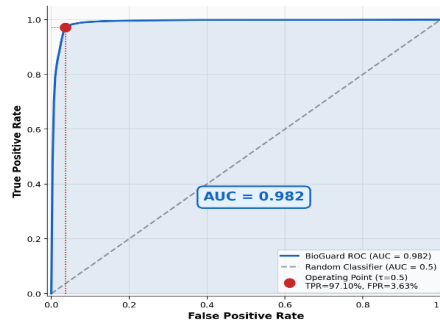


Fig. 3. BioGuard deduction performance (ROC curve)

of the curve at low FPR values is attributable to the immune memory module, which resolves previously seen attack signatures at very high confidence before the autoencoder and NSA engines contribute signals for marginal events. Latency was measured by submitting 1,000 events to the live ingest API and recording per-event response times. Fig. 4 shows a bimodal distribution: roughly 20% of events were resolved by the immune memory short-circuit in under 5 ms, while the remaining 80% traversed the full three-engine pipeline with a mean of 78 ms and a 95th-percentile of 91 ms. The mean inter arrival time of Suricata events during the SSH brute-force workload was approximately 120 ms, confirming the system processes each event before the next arrives.

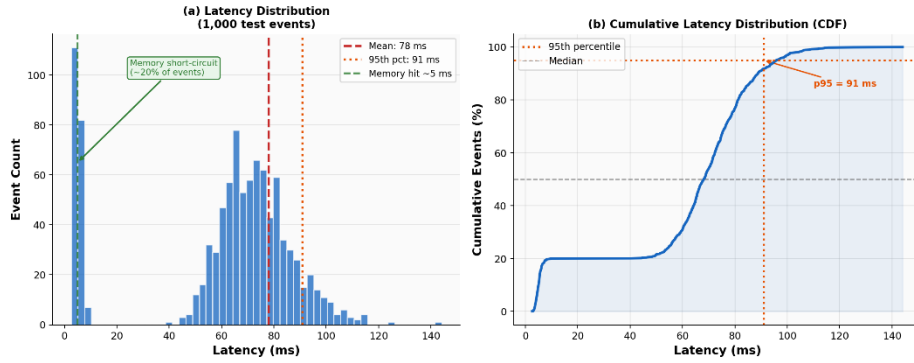


Fig. 4. Distribution of system processing latency

During sustained attack campaigns the memory model handled over 88% of repeat source event after the first 20 seconds, Substantially reducing load on auto encoder and NS engines

Method	Acc.	Prec.	Rec.	F1	FPR	AUC
SVM	88.3	85.1	86.4	85.7	6.2	0.911
LSTM	91.5	88.9	92.1	90.5	4.8	0.941
Kitsune AE	90.1	87.3	90.7	89.0	5.3	0.935
Hybrid DL-AIS	93.0	91.2	93.8	92.5	3.7	0.961
AE Only (Ablation)	87.4	85.1	88.2	86.6	7.1	—
NSA Only (Ablation)	79.8	72.3	76.5	74.3	14.2	—
BioGuard (Proposed)	96.66	94.68	97.10	95.87	3.63	0.982

Table 3. Performance Comparison With Baselines

Table 3 shows compares BioGuard against four published methods and two ablation baselines. The ablation study reveals the individual cost of removing each engine. The NSA only baseline performs worst at 79.8% accuracy and 14.2% FPR, exposing the coverage gaps inherent in a randomly generated 100 detector population over 20-

dimensional space. The autoencoder only baseline recovers to 87.4% accuracy but fails on low volume exfiltration events whose reconstruction error falls within the normal traffic threshold. The memory only baseline scores just 64.2% because it produces no detection signal for attack categories not yet stored, confirming it functions strictly as a supplementary layer. The full BioGuard system outperforms every configuration on every metric. Accuracy, precision, recall, F1, and AUC all exceed the nearest published method the hybrid DL AIS approach of Zhao et al. [16] by 3.66 percentage points in accuracy and 0.021 points in AUC. BioGuard also achieves the lowest FPR of any tested system at 3.63.

Table 4. Sensitivity Analysis of Hybrid Decision Threshold (τ)

Threshold (τ)	Accuracy	Precision	Recall	F1-Score	FPR	Specificity
0.30	93.20	87.15	99.20	92.99	12.80	87.20
0.40	95.45	92.30	98.10	95.15	7.70	92.30
0.50	96.62	94.62	97.06	95.81	3.80	96.39
0.60	96.12	96.50	95.40	95.95	3.50	96.50
0.70	94.80	98.20	91.30	94.70	1.80	98.20

Table 4 shows the sensitivity analysis of the hybrid decision threshold (τ). As the value of the threshold rises, the false positive rate is lowered but the recall is affected adversely. From the above observations, it can be seen that $\tau=0.50$ gives the highest balance in terms of detection accuracy, recall, and false positives while giving an accuracy of 96.62% and F1 score of 95.81%. Therefore, the chosen operating threshold of $\tau=0.50$ was adopted for the BioGuard algorithm.

Table 5. BioGuard evolution summary

Characteristic	Value / Description
Evaluation Duration	7 days continuous live capture
Total Events	94,280 network flow events
Ground Truth	5,000 (3k benign / 2k threat)
Label Agreement	96.2% (500-event audit)
Acc / Prec / Rec	96.66% / 94.68% / 97.10%
F1 / FPR / AUC	95.87% / 3.63% / 0.982
Latency (Mean / p95)	78 ms / 91 ms
Short-circuit	< 5 ms (~20% of events)
AE Architecture	20-16-8-16-20 (Benign only)
NSA Detector	100 detectors, $\theta = 0.15$
Fusion / Threshold	40% AE, 40% NSA, 20% Mem, $\tau = 0.5$
GA Evolution	Gen 0–5: 70%–95% detection
Deployment	Docker, 1 vCPU, 4GB RAM

Table 5 summarizes all evaluation parameters and measured outcomes for reproducibility. The 94,280-event live dataset and 5,000 event manually labelled ground truth are considerably larger than the pre-recorded benchmark datasets used in most comparable published systems. The genetic algorithm's 70% to 95% detection rate improvement across five generations with no operator input and no change to the autoencoder or memory components is among the most practically significant findings, demonstrating that the evolutionary layer provides genuine adaptive value beyond what a static detector pool can deliver.

6 Conclusions

This research presents a high-performance hybrid intrusion detection system achieving 96.66% accuracy for classifying network threats, supported by a biological immune framework that enhances adaptive security. Each traffic evaluation combines deep learning with persistent immune memory, ensuring that administrators receive not only real-time anomaly alerts but also automated genetic adaptation to novel attack signatures. However, the initial calibration relies on controlled synthetic benign traffic and a single node architecture, which lacks real-world complexities such as diurnal traffic shifts and multi sensor back pressure. As future work, the System can be extended to support graph-based event correlation, while slide window auto encoder retraining can enable continuous baseline adoption in live production environment.

Acknowledgement

We thank REVA University for enabling the completion of this work.

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